

Nama Water Services Company SAOC
Consultancy Services for Design and Supervision for STP, Sewer and TE Networks
Systems in Ibri
Tender No. T/2719110/2025

Concept Design Report

Proposed Structure, submitted for approval

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Project 2621 · Revision 0 · August 2026

1 Purpose of this submission

The Terms of Reference require the structure of the Concept Design Report to be approved before the report itself is prepared: “The Consultant shall submit complete concept design report structure for approval according to that will submit comprehensive concept report” (Scope of Work, page 63 of 204).

This document is that submission. It sets out the proposed structure, shows how it maps onto the forty numbered concept deliverables listed in the Scope of Work, and identifies the decisions required from Nama Water Services before the report can be completed.

1.1 Why the structure matters commercially

Provision	Reference
Concept Design is 60 days, plus a separate 21-day review	Tender p147, and as a contract term p177
15 to 20 per cent of the design fee is released on approval of the Concept Design	Tender p153 and p154
Delay is charged at 1 per cent of the Design Stage Contract Value per week	Tender p151, p180, p193
Preliminary Design begins only after STP concept approval	Tender p148 and p178
The concept fee certifies in three separable parts	Bill of Quantities items A1, A2 and A3

The report must therefore be a self-contained, formally approvable document, and its sections must map cleanly onto the three certifiable parts so that each can be signed off independently.

2 Two matters to settle before drafting

2.1 The meaning of TE

PAM-GUD-201 and PAM-GUD-203 define TE as Trade Effluent and TSE as Treated Sewage Effluent. The Terms of Reference, the tender and this project use TE to mean treated effluent. A reader familiar with the guidelines will understand a trade effluent network wherever the report says TE network.

We propose to adopt TSE throughout the report for the treated product of the plant, with an explicit note in the abbreviations, and invite confirmation.

2.2 A cross-reference in the Terms of Reference

The Terms of Reference require the STP location report to follow “item 2.1 in WASTEWATER DESIGN MANUAL Section 05”. PAM-GUD-203 Revision 01 of March 2026 merged the former manuals into a single document and renumbered them; there is no Section 05 and no item 2.1. The equivalent content is now Section 10.1, Site Selection, at pages 63 and 64.

We propose to follow Section 10.1 and invite confirmation.

3 Proposed structure

The third column names the section of Tutorial T03 Revision 01 that sets out the method, the equations and their source pages for that part of the work. A dash indicates a section for which the methodology document does not yet carry a method.

	Section	Method in T03 Rev 01
Front matter		
F1	Document control	
F2	Authorship and discipline signatures	
F3	Abbreviations and definitions	Front matter
F4	Executive summary	
Part A	Project, scope and process	
1	Introduction, background and RG Master Plan context	
2	Scope and boundaries	
3	Programme and design stage gates	
4	Meetings, consultation and decisions record	
5	Quality plan and HSE plan	
Part B	Data	
6	Data collection register	3
7	Data validation and adequacy assessment	3
8	Survey, geotechnical investigation and trial pits	13
9	Existing systems — as-built and GIS	15
Part C	Basis of design	
10	Codes, standards and the deviations register	22
11	Level of service, resilience and contingency	—
12	Design criteria — networks	9, 11
13	Design criteria — STP and process	16
Part D	Demand and flows	
14	Population and land use	3, 4
15	Wastewater generation and design flows	5, 6, 7, 8
16	Trade effluent and industrial contributions	16.4
17	Treated effluent demand and customers	17
Part E	Existing system	
18	Hydraulic assessment of existing networks	14, 15
19	Rehabilitation and upgrading	15
20	RG Master Plan verification	4, 7
Part F	Options	
21	Options methodology	21
22	Sewer network options	9, 10
23	Pumping and lifting stations, force mains, surge	11, 12
24	Treated effluent network options	17
25	STP options — technology, siting, phasing	16
26	Excess effluent, emergency provisions, tankers	16.8
27	Sludge management strategy	18
Part G	Assessment and appraisal	

	Section	Method in T03 Rev 01
28	Ground model and geotechnical report	13
29	Flood protection assessment	16.9
30	Odour assessment and buffer derivation	12, 16.9
31	Climate resilience	—
32	Environmental and social assessment, EIA scoping	—
33	Utility interfaces and the NOC register	13
34	Sustainability — carbon, circular economy, ICV	20.5
35	Cost — CAPEX, OPEX, life cycle cost	19, 20
36	Risk register	20.6
37	Value engineering study	21.5
38	Multi-criteria results and recommendation	21.4
Part H	Delivery	
39	Implementation roadmap	—
40	Contracting strategy and packaging	—
41	Project Integration Plan	—
42	Conclusions and recommendations	
43	Appendices	

4 Presentation conventions

These conventions apply throughout the report and are stated here so that they form part of what is approved.

Element	Convention
Equations	Native Word objects, editable in the equation editor. Each is numbered and followed by a table giving every symbol, its meaning and its unit
Footnotes	Footnotes restart at 1 on each page. They carry the guideline reference and page, the caveat and the derivation; matter belonging to the argument is in the body text
Figures	Maps, process flowcharts and data charts share one numbering sequence. The caption carries the number
Tables	A separate sequence, caption above the table. Header rows repeat across a page break and a row is not split
Maps	Produced from the project geographic information system, with a satellite background, a legend limited to the layers that map draws, and a box giving the quantities behind the figure
Charts	Drawn from the same measured values as the text. Where an asset is recorded as proposed rather than constructed it is shown distinctly and identified in the legend
Revisions	Each issued revision is retained unaltered; a revision is never overwritten

5 The existing network comprises two networks

This is recorded here because it determines what several sections of the report are able to state. The wastewater dataset supplied holds a constructed network and a proposed network, distinguished by the operational status field and confirmed by four further fields that agree with it on every record.

Asset	Constructed	Proposed
Gravity sewer	111.6 km, installed 2006	199.3 km
Force or pumping main	10.0 km, installed 2006	23.2 km
Treated effluent main	none	45.7 km

Neither diameter nor invert level is recorded on the constructed network, and the records carry a remark that the data is to be used for reference only. Section 18, the hydraulic assessment of the existing networks, therefore cannot be completed from the data supplied and waits on the survey now in progress.

6 Sections without a documented method

Six sections of the report have no corresponding method in the current revision of the methodology document. They are listed here so that the gap is visible rather than discovered during drafting.

Section	Why it is outstanding
11 Level of service, resilience and contingency	Requires the failure-mode analysis and criticality framework to be established with NWS
31 Climate resilience	The guideline sets a 50-year horizon at +2 °C and +4 °C; the method for applying it to site selection is not yet written
32 Environmental and social assessment	Depends on the ruling sought on EIA timing — see Section 7
39 Implementation roadmap	Follows from the recommended option, so it is written last
40 Contracting strategy	A separate workshop is required under the Terms of Reference
41 Project Integration Plan	A named deliverable in PAM-GUD-201 page 107; scope to be agreed

None of these is a blocker for the structure itself. Each will be added to the methodology document before the corresponding report section is drafted.

7 Decisions required from Nama Water Services

The following items either block the report or would make it wrong if left unresolved. They are set out in full, with the supporting references, in the accompanying register of decisions and clarifications.

	Item
1	The design horizon. The Inception Report states that planning extends to 2100 as instructed by NWS; the Terms of Reference state completion plus 25 years or ultimate saturated flow
2	Which programme governs, given that the kickoff presentation and the Inception Report carry schedules differing by approximately two weeks throughout
3	The approved project boundary, two versions of which were issued in the same package, differing by 18 per cent against a survey scope priced at approximately 450 km ²
4	Georeferenced versions of Figures 1, 2 and 3, which define the concept and as-built boundaries and are currently supplied as images without coordinates
5	The modelling software, named as MIKE URBAN in the tender staffing schedule and as SewerGEMS and WaterGEMS in the Scope of Work
6	The timing of the Environmental Impact Assessment, placed at scoping in one clause of PAM-GUD-201 and as a full assessment at concept stage in another

Sequencing. Items 1 to 4 determine work that would otherwise be done twice. We recommend they are resolved ahead of the remaining concept programme rather than alongside it.